

$$B \in \mathbb{C}^{P \times L}$$

$$B = [b_1 \ b_2 \ \dots \ b_L]$$

b_k : k -th column of B .

$$Y = \underbrace{[b_{\psi_1} \ b_{\psi_2} \ \dots \ b_{\psi_q}]}_{B_{\psi}} S$$

$$\psi = \{\psi_1, \psi_2, \dots, \psi_q\} \subset \{1, 2, \dots, L\}$$

columns of Y are spanned by columns of B
indexed by ψ

Problem: B and Y are given.

Y is spanned by B_{ψ} but we do not know ψ

We know $|\psi| = q$.

Naive approach: try every possible $\binom{L}{q}$ combinations

$$\binom{L}{q} = \frac{L!}{q!(L-q)!} \approx \left(\frac{eL}{q}\right)^q$$

Assumption: Any $q+1$ columns of B are linearly independent.

$$Y = B_{\psi} S \quad S \text{ has full row rank.}$$

$$\text{span}(Y) = \text{span}(B_{\psi})$$

Find a matrix $Q \in \mathbb{C}^{p \times (L-q)}$

$$Q^H Q = I_{L-q}$$

$$\text{span}(Q) = \text{span}(Y)^\perp$$

$$Y Y^H \xrightarrow[\text{decomposition.}]{\text{eigenvalue}} p \times p$$

$$Y Y^H = \overset{p \times p}{U} \overset{p \times p}{\Lambda} \overset{p \times p}{U}^H$$

$$= \begin{bmatrix} u_1 & u_2 & \dots & u_p \end{bmatrix} \begin{bmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_p \end{bmatrix} \begin{bmatrix} u_1^H \\ u_2^H \\ \vdots \\ u_p^H \end{bmatrix}$$

(The diagonal elements $\lambda_1, \lambda_2, \dots, \lambda_p$ are circled in red, with a red arrow pointing to λ_p and the text "= 0" below it.)

$$\text{span}(Y) = \text{span}(B_{\mathcal{U}}) \quad B_{\mathcal{U}} \in \mathbb{C}^{p \times q}$$

$$\text{rank}(Y) = \text{rank}(B_{\mathcal{U}}) = q \quad p > q$$

$$Q = \begin{bmatrix} u_{q+1} & u_{q+2} & \dots & u_p \end{bmatrix} \quad \lambda_{q+1} = \lambda_{q+2} = \dots = \lambda_p = 0$$

Let b_k be the k th column of B .

$$\text{Test: } Q^H b_k = 0 \Rightarrow k \in \mathcal{U}$$

$$Q^H b_k \neq 0 \Rightarrow k \notin \mathcal{U}$$

If $k \in \mathcal{U}$, then b_k is a column of $B_{\mathcal{U}}$

$$Q^H b_k = 0 \quad \text{because} \quad b_k \in \text{span}(B_{\mathcal{U}})$$

If $k \notin \mathcal{U}$, then $[B_{\mathcal{U}}; b_k] \in \mathbb{C}^{p \times (q+1)}$ has full column rank.
 $b_k \notin \text{span}(B_{\mathcal{U}}) \Rightarrow Q^H b_k \neq 0$

$$A = \begin{bmatrix} B & C \\ C^H & D \end{bmatrix}$$

$$S = D - C^H B^{-1} C \quad \text{Schur complement of } A$$

$$\begin{bmatrix} B & C \\ C^H & D \end{bmatrix} = \begin{bmatrix} I & 0 \\ C^H B^{-1} & I \end{bmatrix} \begin{bmatrix} B & 0 \\ 0 & S \end{bmatrix} \begin{bmatrix} I & B^{-1}C \\ 0 & I \end{bmatrix}$$

$$\begin{bmatrix} B_{\mathcal{U}}^H \\ b_k^H \end{bmatrix} \begin{bmatrix} B_{\mathcal{U}} & b_k \end{bmatrix} = \begin{bmatrix} B_{\mathcal{U}}^H B_{\mathcal{U}} & B_{\mathcal{U}}^H b_k \\ b_k^H B_{\mathcal{U}} & b_k^H b_k \end{bmatrix} = A$$

$$S = b_k^H b_k - b_k^H B_{\mathcal{U}} (B_{\mathcal{U}}^H B_{\mathcal{U}})^{-1} B_{\mathcal{U}}^H b_k$$

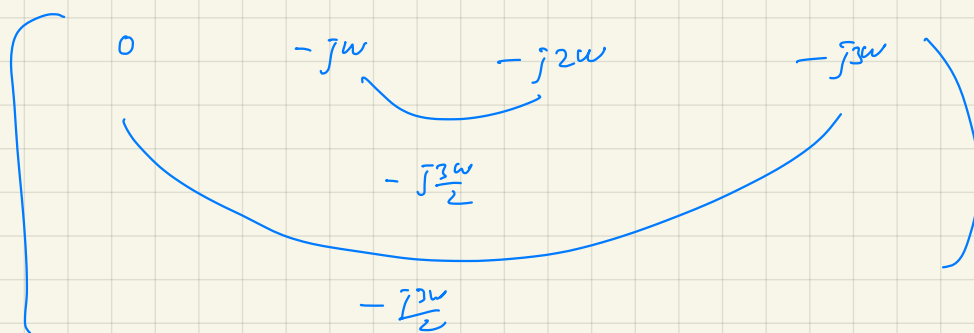
$$= b_k^H \underbrace{\left(I - B_{\mathcal{U}} (B_{\mathcal{U}}^H B_{\mathcal{U}})^{-1} B_{\mathcal{U}}^H \right)}_{Q Q^H} b_k = b_k^H Q Q^H b_k$$

$$= \|Q^H b_k\|_2^2 \geq \lambda_{\min}(A)$$

$$h[n] = f[n] + 2f[n-1] + 2f[n-2] + f[n-3]$$

compute $H(e^{j\omega}) = A(\omega) e^{j\phi(\omega)}$

$$H(e^{j\omega}) = 1 + 2e^{-j\omega} + 2e^{-j2\omega} + e^{-j3\omega}$$



$$H(e^{j\omega}) = e^{-j\frac{3\omega}{2}} \left(e^{j\frac{3\omega}{2}} + 2e^{j\frac{\omega}{2}} + 2e^{-j\frac{\omega}{2}} + e^{-j\frac{3\omega}{2}} \right)$$

$$= \underbrace{e^{-j\frac{3\omega}{2}}}_{\text{phase}} \underbrace{\left[\cos\left(\frac{3\omega}{2}\right) + 2\cos\left(\frac{\omega}{2}\right) \right]}_{\substack{\in \mathbb{R} \\ A(\omega)}}$$

$$\phi(\omega) = -\frac{3\omega}{2}$$

$$-\frac{3}{2} \cdot 2 \in \mathbb{Z}$$

GLP

Type-I GLP filter

- $L = \text{length of filter}$ is an odd #
- $h[n]$ is real-valued.

claim GLP $\phi(\omega) = -\frac{L-1}{2}\omega \iff h[n] = h[L-1-n] \quad \forall n$

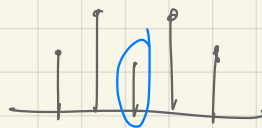
ex) $L=5$



proof ($\Leftarrow$)

$$H(e^{j\omega}) = \sum_{n=0}^{L-1} h[n] e^{-j\omega n}$$

$$= \sum_{n=0}^{\frac{L-1}{2}-1} h[n] e^{-j\omega n} + h[\frac{L-1}{2}] e^{-j\omega \frac{L-1}{2}} + \sum_{n=\frac{L-1}{2}+1}^{L-1} h[n] e^{-j\omega n}$$



change of variable $m = L-1-n$

$$\sum_{n=\frac{L-1}{2}+1}^{L-1} h[n] e^{-j\omega n} = \sum_{m=0}^{\frac{L-1}{2}-1} h[L-1-m] e^{-j\omega (L-1-m)}$$

$$H(e^{j\omega}) = \sum_{n=0}^{\frac{L-1}{2}-1} h[n] e^{-j\omega n} + h[\frac{L-1}{2}] e^{-j\omega \frac{L-1}{2}} + \sum_{n=0}^{\frac{L-1}{2}-1} \underbrace{h[L-1-n]}_{h[n]} e^{-j\omega(L-1-n)}$$

$$= h[\frac{L-1}{2}] e^{-j\omega \cdot \frac{L-1}{2}} + \sum_{n=0}^{\frac{L-1}{2}-1} h[n] \left[e^{-j\omega n} + e^{-j\omega(L-1-n)} \right]$$

$$n + (L-1-n) = L-1$$

$$= h[\frac{L-1}{2}] e^{-j\omega \frac{L-1}{2}} + \sum_{n=0}^{\frac{L-1}{2}-1} h[n] e^{-j\omega \frac{L-1}{2}} \left[\underbrace{e^{-j\omega(n - \frac{L-1}{2})} + e^{j\omega(n - \frac{L-1}{2})}}_{2 \cos(n - \frac{L-1}{2})} \right]$$

$$= e^{-j\omega \frac{L-1}{2}} \left[h[\frac{L-1}{2}] + \sum_{n=0}^{\frac{L-1}{2}-1} 2 h[n] \cos(n - \frac{L-1}{2}) \right]$$

$$A(\omega)$$

$$\phi(\omega) = -\frac{L-1}{2} \omega$$

($\Rightarrow$) Assume $H(e^{j\omega}) = A(\omega) e^{-j\omega \frac{L-1}{2}}$

goal: $h[L-1-n] = h[n]$

$$h[L-1-n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} H(e^{j\omega}) e^{j\omega(L-1-n)} d\omega$$

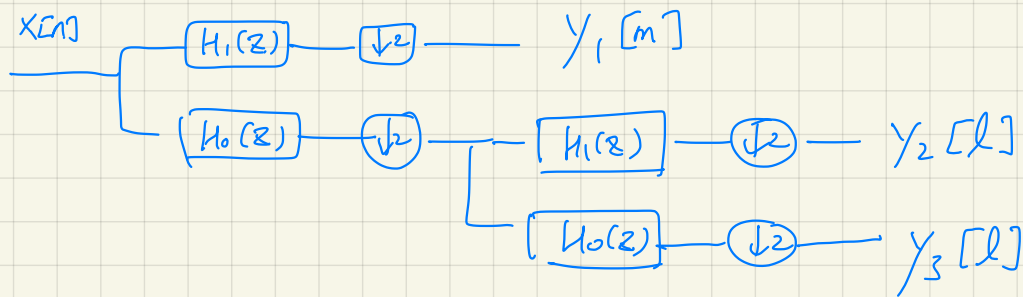
$$= \frac{1}{2\pi} \int_{-\pi}^{\pi} A(\omega) e^{-j\omega \frac{L-1}{2}} e^{j\omega(L-1-n)} d\omega$$

$$= \frac{1}{2\pi} \int_{-\pi}^{\pi} A(\omega) e^{j\omega \frac{L-1}{2}} e^{-j\omega n} d\omega$$

$$= \left(\frac{1}{2\pi} \int_{-\pi}^{\pi} \underbrace{A(\omega) e^{-j\omega \frac{L-1}{2}}}_{H(e^{j\omega})} e^{j\omega n} d\omega \right)^*$$

$\underbrace{\hspace{10em}}_{h[n]}$

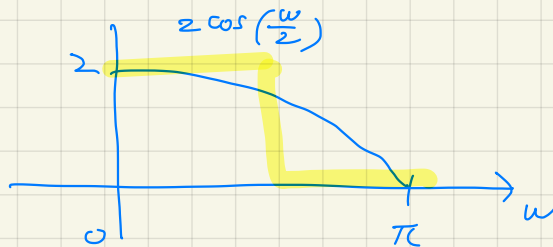
$$= h[n]$$



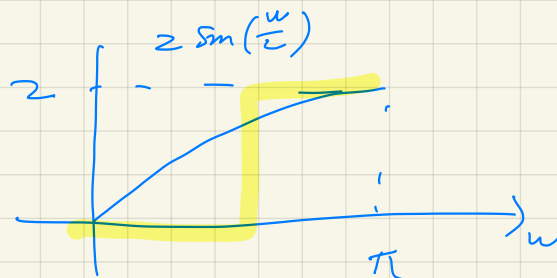
$$H_1(z) = 1 - z^{-1} \quad H_0(z) = 1 + z^{-1}$$

Draw $|H_1(e^{j\omega})|$ & $|H_0(e^{j\omega})|$

$$\begin{aligned} H_0(e^{j\omega}) &= 1 + e^{-j\omega} = e^{-j\frac{\omega}{2}} (e^{j\frac{\omega}{2}} + e^{-j\frac{\omega}{2}}) \\ &= e^{-j\frac{\omega}{2}} 2 \cos\left(\frac{\omega}{2}\right) \end{aligned}$$



$$\begin{aligned} H_1(e^{j\omega}) &= 1 - e^{-j\omega} = e^{-j\frac{\omega}{2}} (e^{j\frac{\omega}{2}} - e^{-j\frac{\omega}{2}}) \\ &= e^{-j\frac{\omega}{2}} 2j \sin\left(\frac{\omega}{2}\right) \\ &= e^{-j(\frac{\omega}{2} + \frac{\pi}{2})} 2 \sin\left(\frac{\omega}{2}\right) \end{aligned}$$



Equivalent system?

